

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Basics of COA

Lecture No.- 01



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Topics to be Covered



Vishradeep sir PW



Topic

Prerequisites

Topic

Why COA

Topic

Data In Computers

Topic

Components of Computer

Topic

Binary Numbers



Topic : Introduction

❑ **GATE Ranks:**

- 682 (2009) – 3rd year
- 19 (2010) – 4th year
- 119,440 etc.

❑ **Education:**

- ME from IISc Bangalore
- M. tech from BITS-pilani in Data Science

❑ **Work:**

- 18+ Year Teaching Experience
- 14+ in GATE/IES
- Worked in Software Industry

CoA



dynamic course

weightage $\Rightarrow$ 8-10 marks

Ranker maker



Topic : Prerequisites

- Basic components of computer: CPU, memory (RAM, ROM, HDD), I/O
- Number system: Binary, Decimal, Hexadecimal etc.
- Digital logic basics: Mux, Decoder etc.

Powers of 2:

Unit	Time	Bit or Byte
K (Kilo)	10^3	$2^{10} = 1024$
M (Mega)	10^6	2^{20}
G (Giga)	10^9	2^{30}
T (Tera)	10^{12}	2^{40}

or addresses

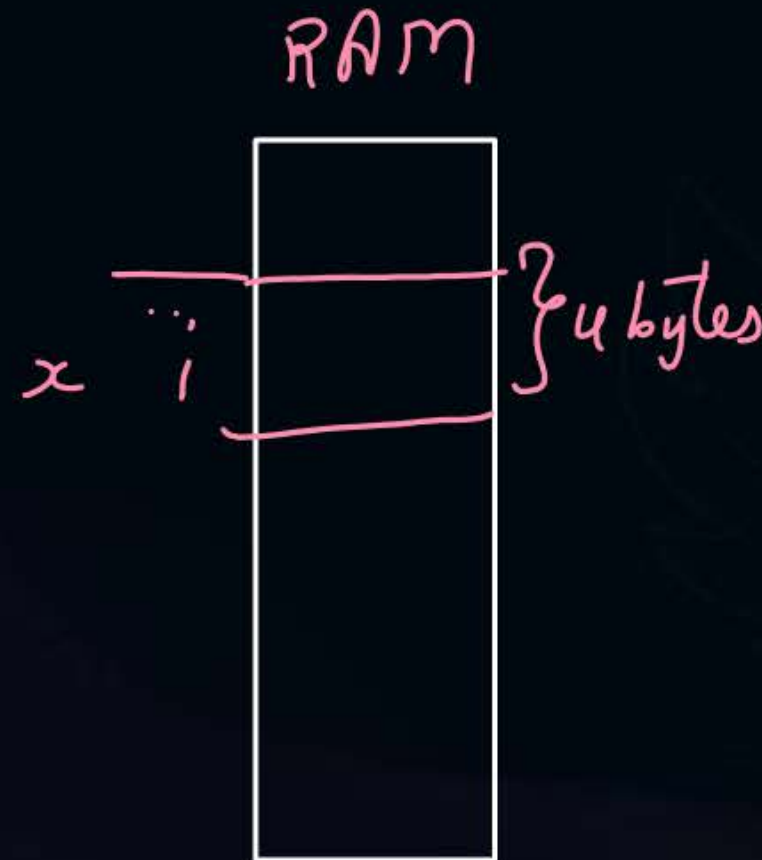
millisec. = 10^{-3} sec
microsec. = 10^{-6} sec
nanosec. = 10^{-9} sec



Topic : Why COA

- To understand: How a computer works
- To understand other courses: OS, Compiler, Programming etc.
- Help in real world development: DBMS, Hardware Design, IoT problems etc.

`int x;`



Syllabus

- Basics of Computer system
- micro-programming
- Instruction & Addressing modes
- CPU & control unit
- *floating point representation
- I/O organization
- mem. —||—
- *Cache
- Disk
- *pipeline

Comp.
arch.

Comp.
org.



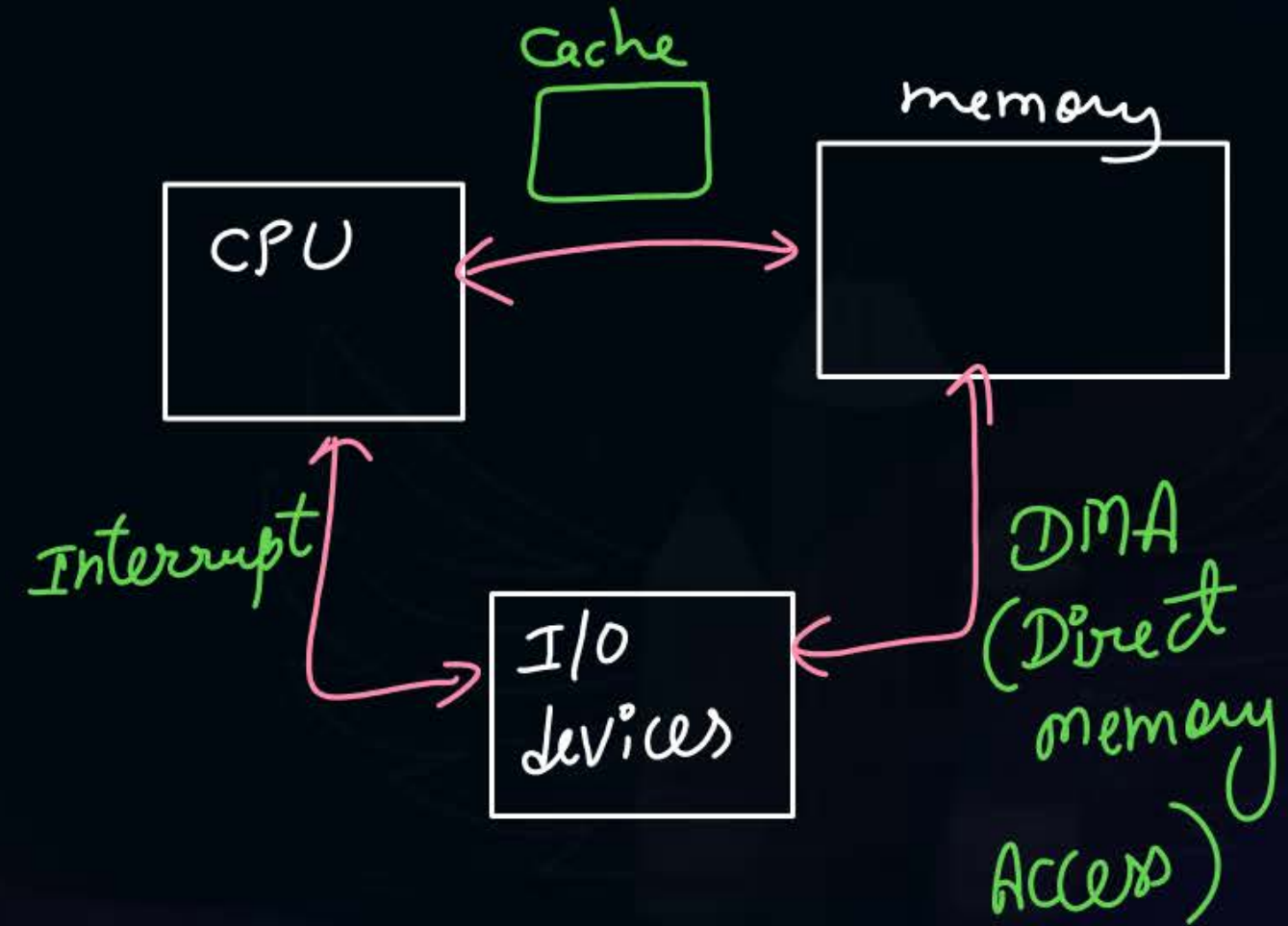
Topic : Computer Organization & Architecture

Computer Architecture:

→ Implementation → Comp. organization

- Conceptual design and fundamental operational structure.

→ what devices in your computer
→ what those devices do
→ CPU design
→ Instructions
→ Addressing modes
→ Data format





Topic : Computer Organization & Architecture

Computer Organization:

- Deals with physical devices and their interconnections
- With a perspective of improving the performance.

How they interact with each other

→ I/O organization
memory organization



Topic : Computer Organization & Architecture

Computer Architecture	Computer Organization
• CPU Design	• I/O Organization
• Instructions	• Memory Organization
• Addressing modes	• Performance
• Data format	



Topic : Data In Computers

Data

Number

Fixed point representation

Floating point representation

unsigned

$5 \Rightarrow 101$
 $8 \Rightarrow 1000$

signed

→ signed magnitude
→ 1^s complement
→ 2^s complement

Prerequisite

character

→ ASCII (American Standard Code for Information Interchange)

→ EBCDIC (Extended Binary Coded Decimal Interchange Code)



Topic : Binary Numbers



$$2^0 = 1$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$2^5 = 32$$

$$2^6 = 64$$

$$2^7 = 128$$

$$2^8 = 256$$

$$2^9 = 512$$

$$2^{10} = 1024$$

$$2^{11} = 2048$$

$$2^{12} = 4096$$

$$2^{13} = 8192$$

$$2^{14} = 16384$$

$$2^x * 2^y = 2^{x+y}$$

$$\frac{2^x}{2^y} = 2^{x-y}$$

bit $\Rightarrow$ single binary digit 0 or 1
byte $\Rightarrow$ combination of 8 bits (ex:- 10101101)

bit $\Rightarrow$ b
byte $\Rightarrow$ B



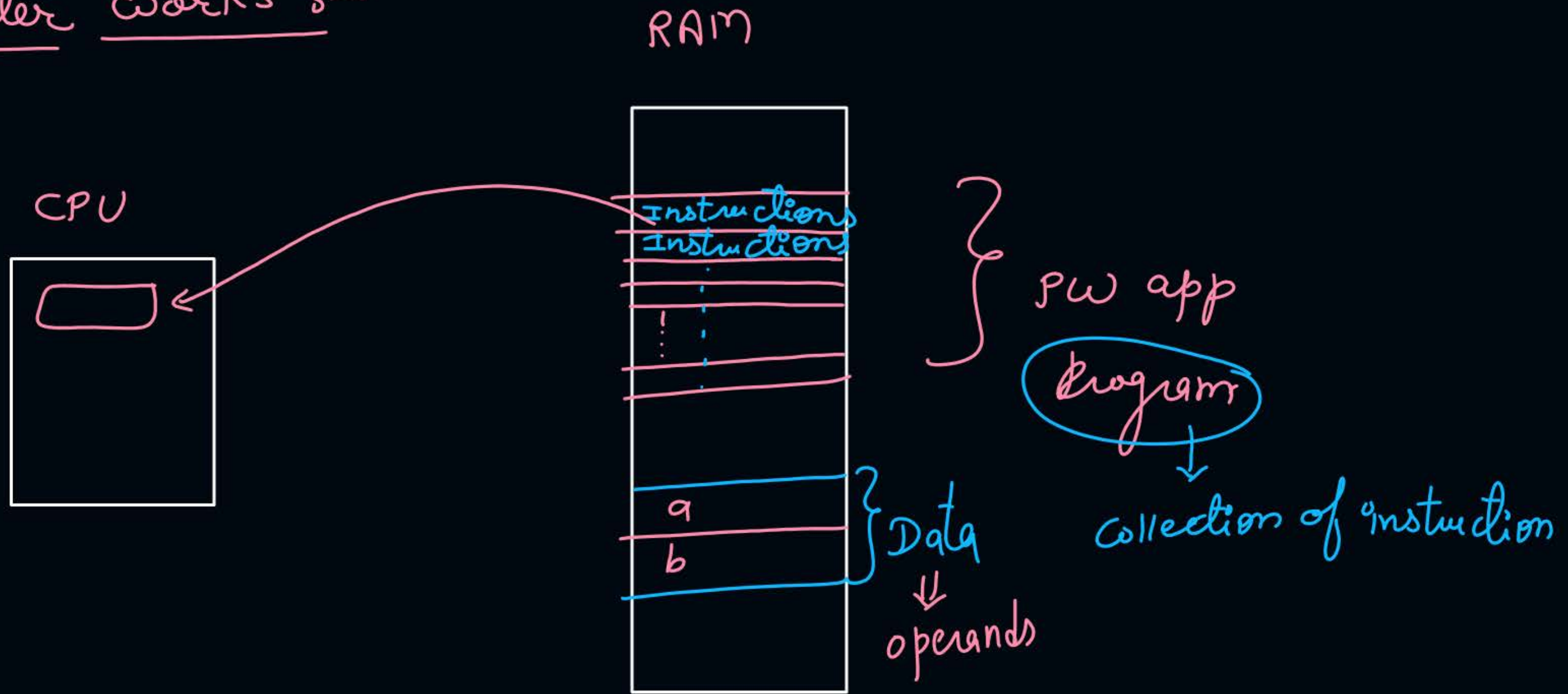
Topic : Components of Computer

- CPU :- Parts
 - C.U. (Control Unit)
 - A.L.U. (Arithmetic Logic Unit)

- Memory: RAM
ROM

- I/O Devices:
 - Input/output
 - Input devices
 - output "
 - storage "

How Computer works :-





Topic : Other Components

- System Buses → A way to connect CPU with memory and I/O devices
- CPU Registers

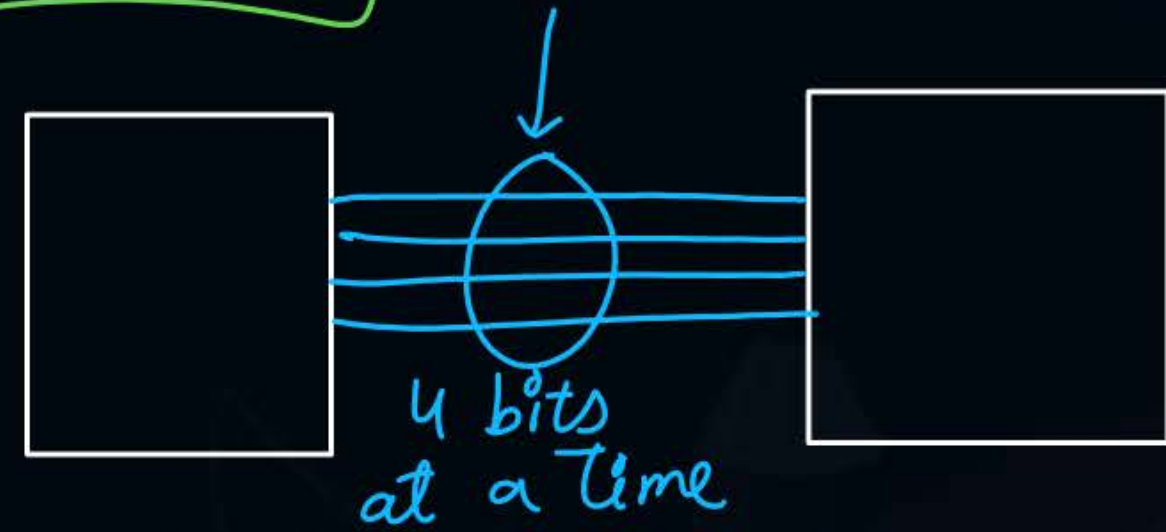
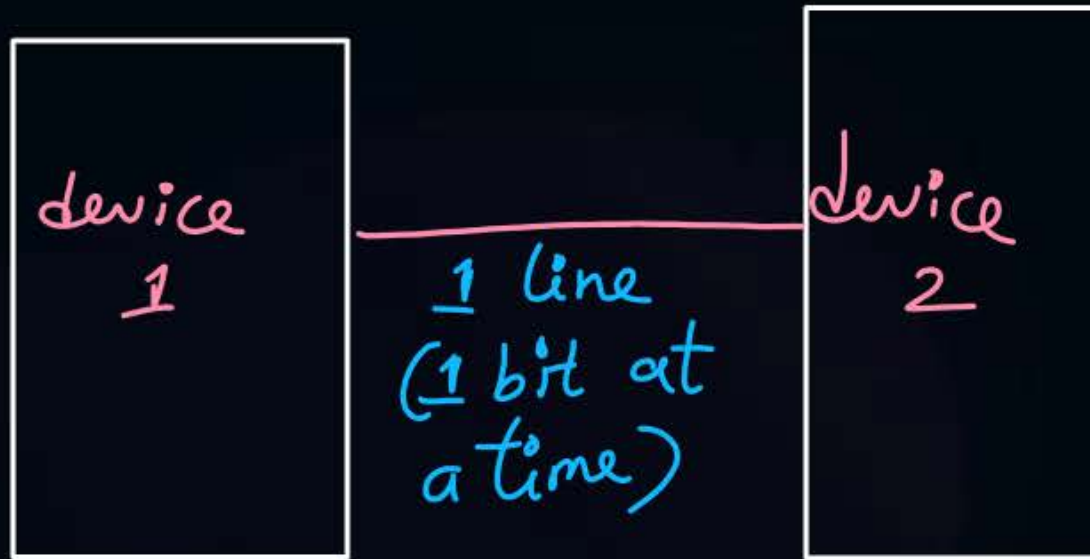


Topic : Other Components

System Buses: collection of communication line to connect CPU with other devices.

example:-

4 lines $\Rightarrow$ a bus

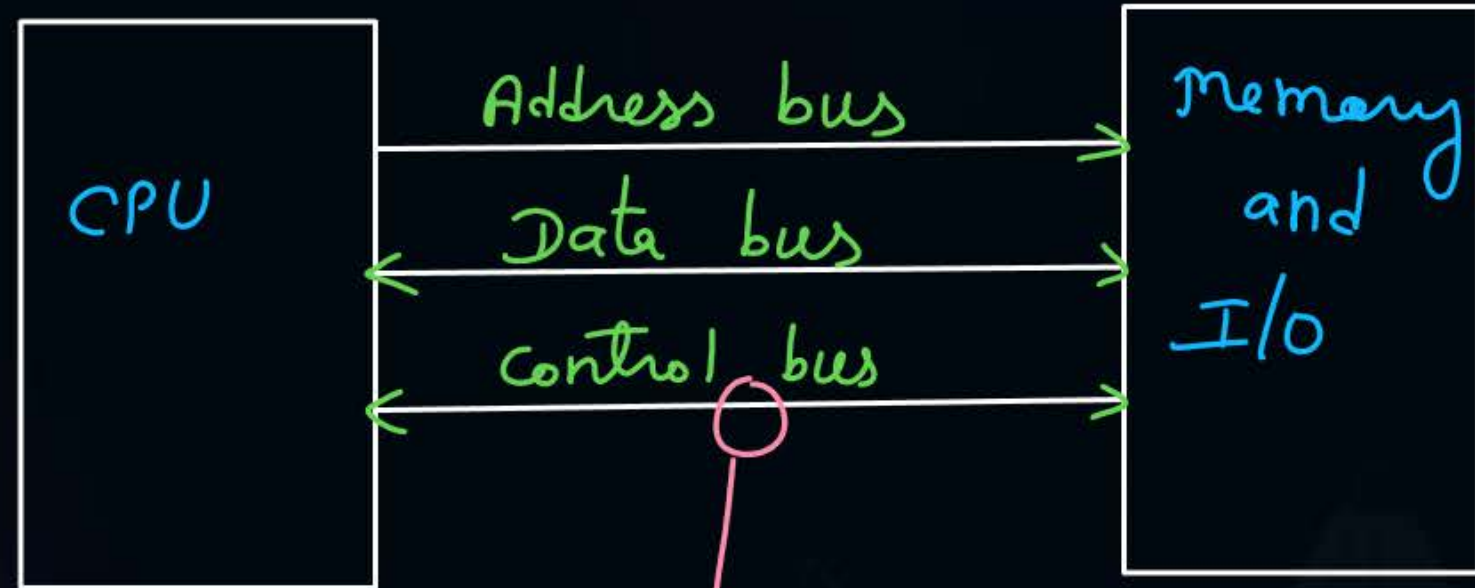




Topic : Other Components

System Buses:

- Address Bus
- Data Bus
- Control Bus



every individual line is unidirectional



Control bus:-

cpu to memory:-

memory Read line
memory write line

from CPU to I/O:-

I/O Read line
I/O write line

memory to CPU:-

Ready
wait

I/O to CPU:-

Interrupt



Prerequisites

- **Number System**

Binary, Hexadecimal, Decimal

Conversion from one system to another

Signed numbers: Sign-Magnitude, 1's Complement, 2's Complement

- **Decoder**

- **Multiplexer**

How to learn CoA:-

→ watch lecture everyday $\Rightarrow$ revise + make notes

→ chapter ends $\Rightarrow$ dpp, PYQs

→ weekend $\Rightarrow$ saturday $\Rightarrow$ revise 1 week topics
sunday $\Rightarrow$ weekly test

subject done $\Rightarrow$ 1 full revision



2 mins Summary



Topic

Architecure vs Organization

Topic

Numbers & Data in Computers

Topic

Components of Computer

Topic

System Buses

Topic

Types of Buses



Happy Learning

THANK - YOU